

Modeling How Macroeconomic Shocks Affect Regional Employment: Analyzing the Brazilian Formal Labor Market using the Global VAR Approach.

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Talking points

- 1 Motivation
- 2 Challenges and Methods
- 3 Data
- 4 Our Results
- 5 Limitations and Possible Extensions
- 6 Conclusions

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Macroeconomics and Regional Data:

- Macroeconomic shocks affect sectors and regions with different intensities and delays;
- Population and economic activities tend to cluster in specific areas due economies of scale and agglomeration.
- The population tends to locate near the sea coast compared with interior continental areas.
- Sector activities tend to be concentrated in some areas or cities.

Goals

- We want to model possible interaction between macroeconomic shocks and regional employment dynamics;
- Assess possible different responses of regional employment to the same shock;
- Generate scenarios for regional employment evolution in Brazil;

Possible innovation

- Apply GVAR to link macroeconomic dimension with regional dimension in Brazil;
 - Deal with curse of dimensionality;
- Use of IBGE Survey on economic links of Brazilian economy to construct weights;

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Global VAR approach

- Global autoregressive vector (GVAR) methodology was made by Pesaran, Weiner and Schuermann (2004), and improved on by Dees, Mauro, Pesaran and Smith (2007);
- Chudik and Pesaran (2011) - Analyzed the curse of dimensionality in the case of large dynamic linear systems.
- Infinite-dimensional VAR could be well approximated by a set of finite-dimensional small-scale models that can be consistently estimated separately.
- Spatial analysis is a natural option on this framework;

The Global VAR model as a two-step procedure

- First stage: Specific models of each region are estimated conditioned to external influences of other regions.

$$x_{it} = a_{i0} + a_{i1}t + \sum_{j=1}^{p_i} \Phi_{ij}x_{i,t-j} + \Lambda_{i0}x_{it}^* + \sum_{l=1}^{q_i} \Lambda_{il}x_{i,t-l} + u_{it}$$

$$x_{it}^* = \sum_{j=0}^N W_{ij}x_{jt}$$

$$A_{i0}W_ix_t = a_{i0} + a_{i1}t + \sum_{l=1}^p A_{il}W_ix_{t-l} + \varepsilon_t$$

The Global VAR model as a two-step procedure

- Second Stage: The VAR models of each region are grouped into a single Global VAR model.

$$G_0 = \begin{bmatrix} A_{00}W_0 \\ A_{10}W_1 \\ \vdots \\ A_{N0}W_N \end{bmatrix}; \quad G_l = \begin{bmatrix} A_{0l}W_0 \\ A_{1l}W_1 \\ \vdots \\ A_{Nl}W_N \end{bmatrix}; \quad a_0 = \begin{bmatrix} a_{00} \\ a_{10} \\ \vdots \\ a_{N0} \end{bmatrix}; \quad a_1 = \begin{bmatrix} a_{01} \\ a_{11} \\ \vdots \\ a_{N1} \end{bmatrix}; \quad \varepsilon_t = \begin{bmatrix} \varepsilon_{0t} \\ \varepsilon_{1t} \\ \vdots \\ \varepsilon_{Nt} \end{bmatrix};$$

$$G_0 x_t = a_0 + a_1 t + \sum_{l=1}^p G_l x_{t-l} + \varepsilon_t$$

$$F_i = G_0^{-1} G_i; \quad b_0 = G_0^{-1} a_0; \quad b_1 = G_0^{-1} a_1; \quad e_t = G_0^{-1} \varepsilon_t;$$

$$x_t = b_0 + b_1 t + \sum F_i x_{t-i} + e_t$$

Selected papers on the subject

- Oliveira and Carneiro (2001) - Analysis of the fluctuations in the employment level of several Brazilian states in relation to national employment with the objective of verifying a long term relationship.
- Camila Kraide Kretzmann and Marina Silva da Cunha (2009) - Analysis of the formal labor market between metropolitan and non-metropolitan areas in Brazil.
- Norbert Schanne (2015) - Analysis of the regional labor market in Germany, focusing on forecasting regional labor markets. She used the GVAR methodology to perform a prospective analysis of the German economy.

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Brazilian Economy Structure



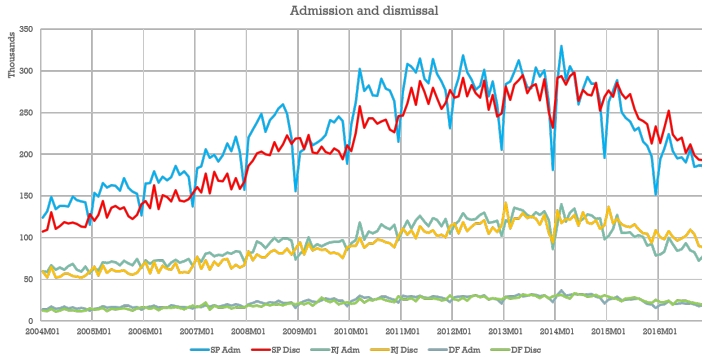
- South and South East - Industrial, services and most developed part of the country
- North East - Agricultural and tourism
- North - Agricultural - Amazon forest;
- Central (0.06) - New agricultural frontier;

Figure: Brazilian Regions

Data

Variable	Source	Frequency	Range
Brazilian Industrial Production	IBGE	Monthly	2004-01 to 2016-12
Admissions and Dismissals	CAGED	Monthly	2004-01 to 2016-12
Meso-regions and municipalities	IBGE	N/A	as of 2015
Connections between regions	IBGE	N/A	as of 2007

Some Graphics



Model

- The units in the model are the mesoregions defined by the Brazilian Institute of Geography and Statistics in 2015.
- The labor market variables are the numbers workers admitted and dismissed from the labor market.

$$x_{it} = \begin{bmatrix} Adm_{it} \\ Dis_{it} \end{bmatrix}; \quad x_{it}^* = \begin{bmatrix} Adm_{it}^* \\ Dis_{it}^* \end{bmatrix}$$

$$Adm_{it}^* = \sum_{j=1}^N W_{ij} Adm_{jt} \quad Dis_{it}^* = \sum_{j=1}^N W_{ij} Dis_{jt}$$

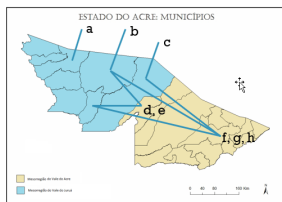
Simple Macroeconomic Model

- The Macroeconomic model contains Brazilian production industrial index and employment admissions and dismissals with feedback from regions:

$$\omega_t = [\ln(IndProd)]$$

$$\Delta \omega_t = \Phi_{\omega 0} + \sum_{l=1}^{k_1} \Phi_{\omega l} \Delta \omega_{t-l} + \sum_{l=1}^{k_2} \Lambda_{\omega l} \Delta x_{i,t-l}^* + \eta_{\omega t}$$

Our weight matrix choice



- Survey of Brazilian Bureau of Geographic and Statistics on economic links of regions;
- Links available at all municipalities;

Figure: Response to changes in growth of industrial production

$$W_{blue} = \frac{\begin{bmatrix} c_{i1} & \dots & c_{iN} \end{bmatrix}}{\sum_{j=1}^N c_{ij}} \quad W_{blue} = \frac{\begin{bmatrix} 5 & 3 \end{bmatrix}}{8} = \begin{bmatrix} 0.625 & 0.375 \end{bmatrix}$$

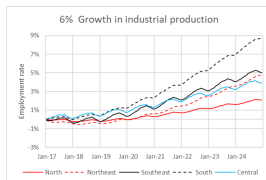
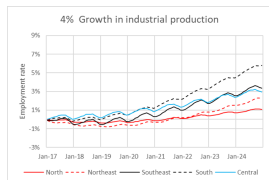
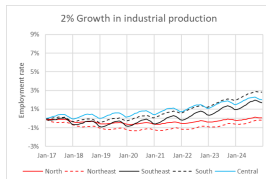
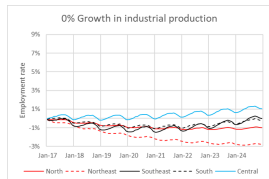
Results

- 137 specific models were specified.
 - Domestic variables: 98 models with lag 2 and 39 models with lag 1
 - Foreign variables: 91 models with lag 2 and 46 models with lag 1.
 - 9 models with 1 cointegration relation and 128 models with 2 cointegration relations.
- Weak exogeneity status are satisfied for most of units;

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Brazilian Five Regions Response



- South (0.18)
- North East (0.16)
- South East (0.1)
- North (0.06)
- Central (0.06);

Table: Response to changes in growth of industrial production

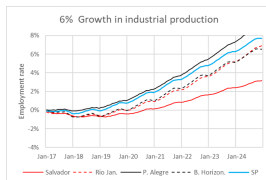
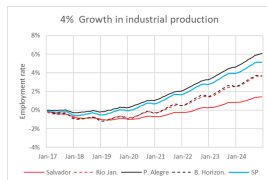
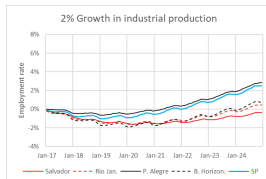
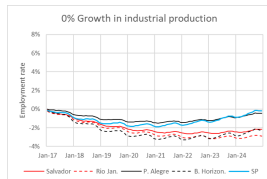
Southeast States Response



- Rio de Janeiro (0.19)
- Espírito Santo (0.09)
- São Paulo (0.08)
- Minas Gerais (0.07);

Table: Response to changes in growth of industrial production

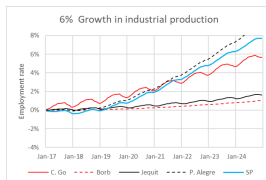
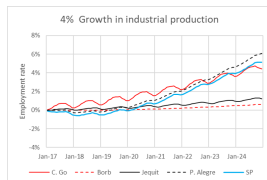
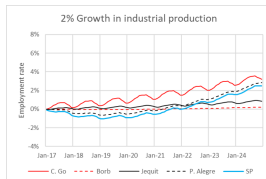
Selected meso regions



- Rio de Janeiro (0.19)
- Porto Alegre (0.19)
- Belo Horizonte (0.17)
- São Paulo (0.16)
- C. Goiás (0.07)
- Salvador (0.11)

Table: Response to changes in growth of industrial production

Selected Mesoregions

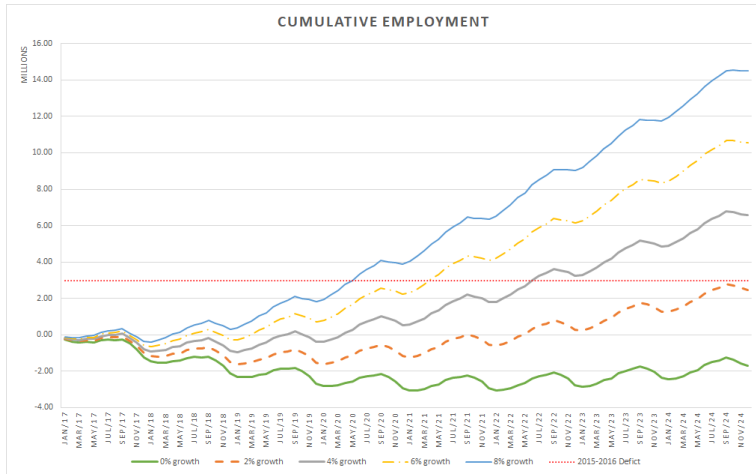


- Porto Alegre (0.19)
- São Paulo (0.16)
- C. Goiás (0.07)
- Borborema (0.02)
- Jequit. (0.02);

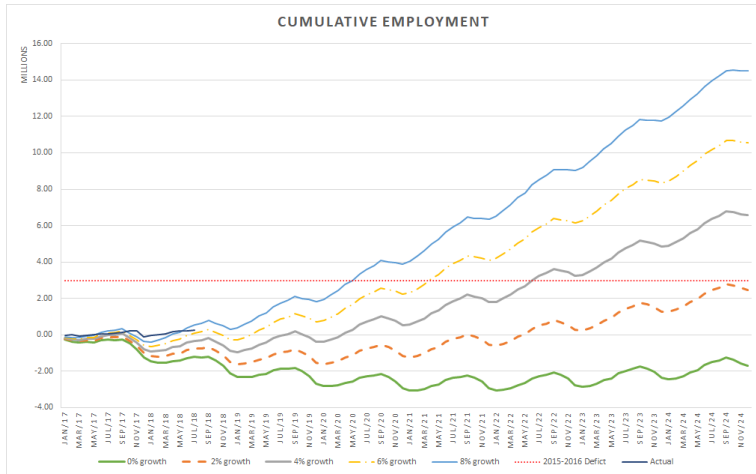
Table: Response to changes in growth of industrial production

- VIDEO

Conditional predictions from different scenarios



Actual vs. Fitted



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Limitations

- Macroeconomic model can be improved to incorporate more variables and disentangle possible different types of shocks;
- Different weights strategies can be tried and compared;
- Evaluation of forecast performance.

Extensions

- Work at higher disaggregated level but “municipality split epidemia” in Brazil must be addressed - Common minimum areas;
- Models can be improved for each unity using *Autometrics* algorithm similar to Ericsson et al. (2012);

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Some tentative conclusions

- South, Southeast and Northeast have shown to respond stronger to shocks in Brazilian aggregate industrial production;
- Macro factor does not play strong influence on North and Central regions meso regions;
- Formal employment will tend to recover slowly in Brazil in near future unless high growth rates starts to be observed;